

TEXT MINING AND SEARCH: TOPIC MODELLING TEXT SUMMARISATION

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SUMMARY

- This project aims to **generate short summaries** of the documents contained in the Wikipedia Featured Articles corpus.
- We perform **topic modelling** using pLSA, LDA and BERTopic, then **text summarisation** through global and local pLSA, LED and LLM.
- Results show that the most coherent topics are obtained from **pLSA**. Regarding text summarisation, **LLM** generates the highest quality summaries, however **global pLSA** appears to be the most reliable technique in terms of factual consistency.



THE DATASET

- The English subset of the “**Wikipedia Featured Articles**” dataset consists of a training set of **3793** documents and a test set of **30** documents.
- We picked this dataset because its **featured articles** grant high linguistic quality, cover a broad range of topics, and come with a human-made summary.
- After loading the dataset, we clean and parse it, then perform **tokenization** and generate **bigrams** and **trigrams** using *gensim*.



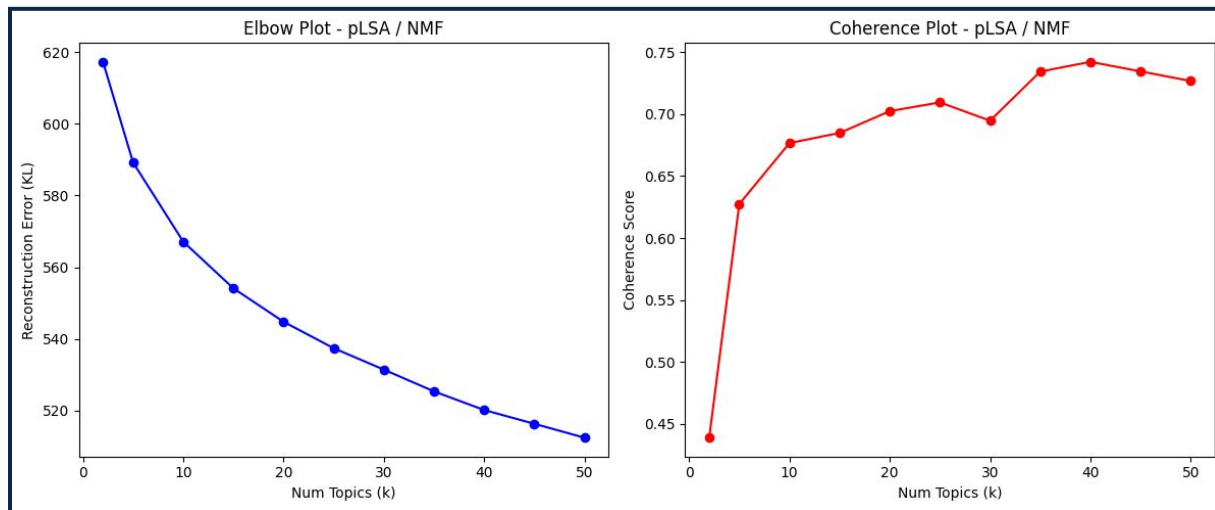
TOPIC MODELLING

- We perform **topic modelling**, looking for latent features (topics) within the corpus.
- After obtaining a coherent and plausible list of topics, we can then use it to evaluate the importance of sentences during text summarisation.
- We will confront **pLSA**, **LDA** and **BERTTopic**, using coherence C_v as topic evaluation metric.



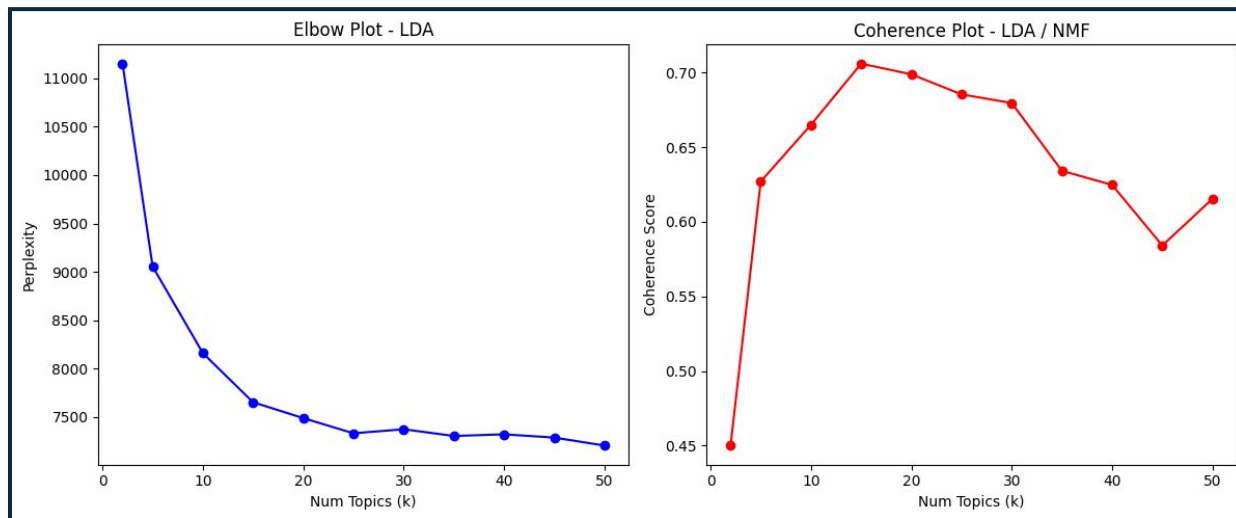
pLSA

- This technique uses a **probabilistic framework**, defining topics as latent variables, using **NMF** to decompose the **co-occurrence matrix $\mathbf{X}$** into a **document-topic matrix $\mathbf{W}$** and a **topic-word matrix $\mathbf{H}$** and applying the **KL-divergence** objective function, we have the mathematical equivalence of pLSA.



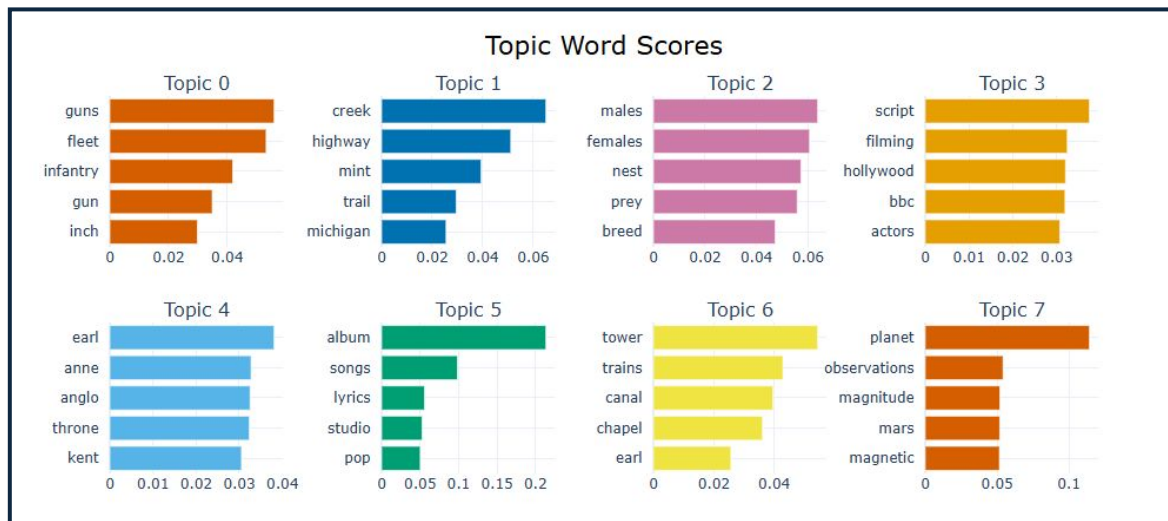
LDA

- LDA follows a probabilistic approach as well, considering documents as a **mixture** of topics and topics as a mixture of words. It utilises **Dirichlet priors** to control the sparsity of topic distributions.



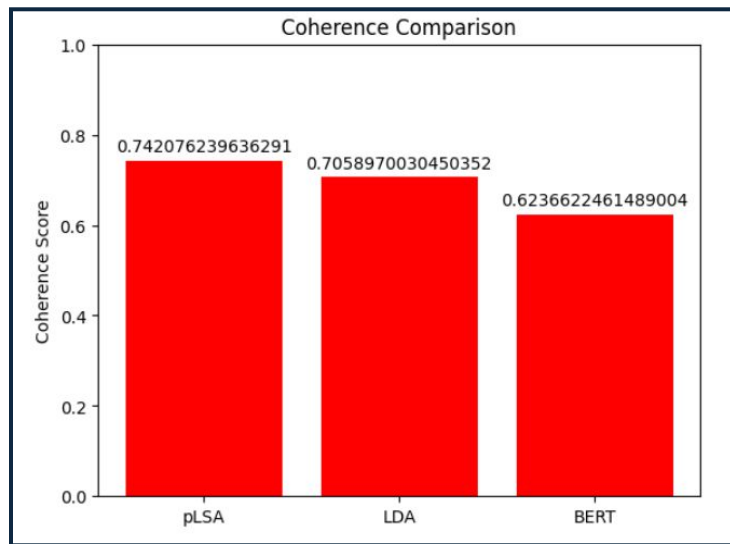
BERTopic

- BERTopic utilises semantic embeddings, clustering them to form topics. Its approach is based on semantic similarity rather than frequency counts. We used *jina-embeddings-v2-small-en*.



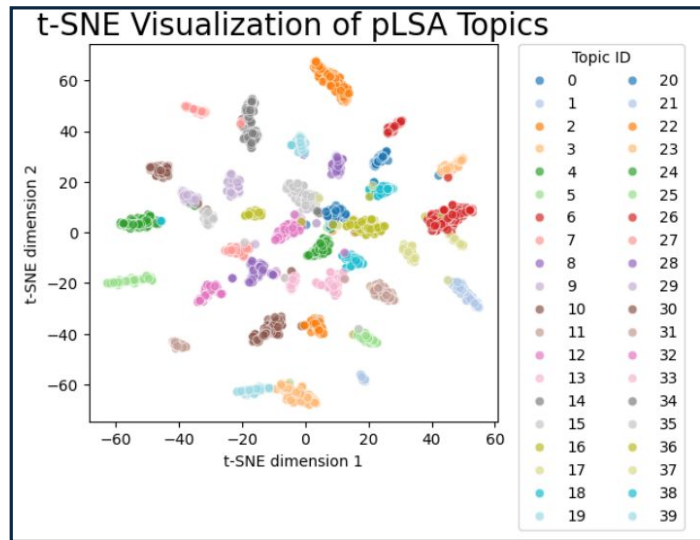
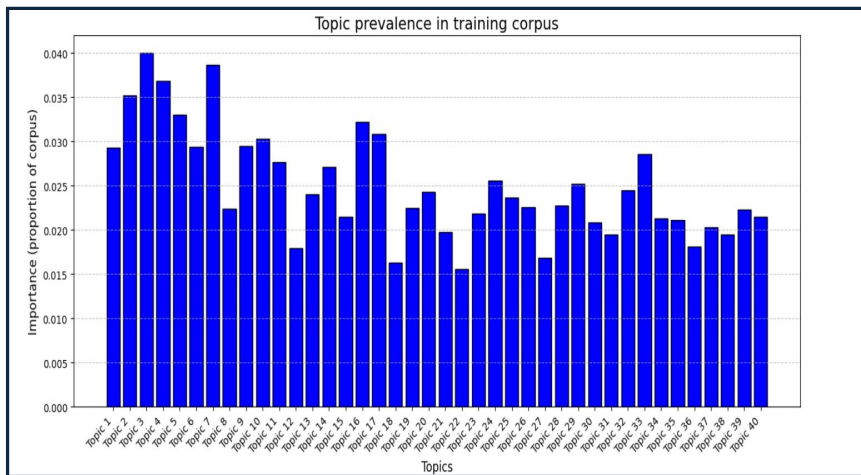
EVALUATION

- After fine tuning the parameters, we evaluate the models computing their **Topic Coherence** score, which measures the semantic similarity between the top words in a topic.
- The best model results to be **pLSA**, achieving a coherence value of **0.7421** with **40 topics**.



RESULTS OF TOPIC MODELLING

- Looking at the topics we obtained, they all appear relatively close in importance. Some of them overlap, but these results are overall **satisfying** and will be useful during text summarisation.



RESULTS OF TOPIC MODELLING

- These are the most important words of the first topics:

Topic 1:

child: 0.3458
family: 0.3432
woman: 0.3337
father: 0.3037
mother: 0.2713
wrote: 0.2620
life: 0.2617
wife: 0.2425
death: 0.2410
friend: 0.2334

Topic 2:

game: 2.0321
player: 0.7772
character: 0.4085
gameplay: 0.2908
video_game: 0.2342
nintendo: 0.2004
released: 0.1896
release: 0.1844
graphic: 0.1740
feature: 0.1606

Topic 3:

album: 1.8446
song: 1.3640
band: 1.2712
music: 0.4782
lyric: 0.3373
recording: 0.3177
record: 0.3165
track: 0.3142
tour: 0.3098
sound: 0.2911

Topic 4:

specie: 1.0513
bird: 1.0281
nest: 0.5331
male: 0.5149
female: 0.4346
subspecies: 0.3335
egg: 0.2933
population: 0.2901
feather: 0.2323
breeding: 0.2258



TEXT SUMMARISATION

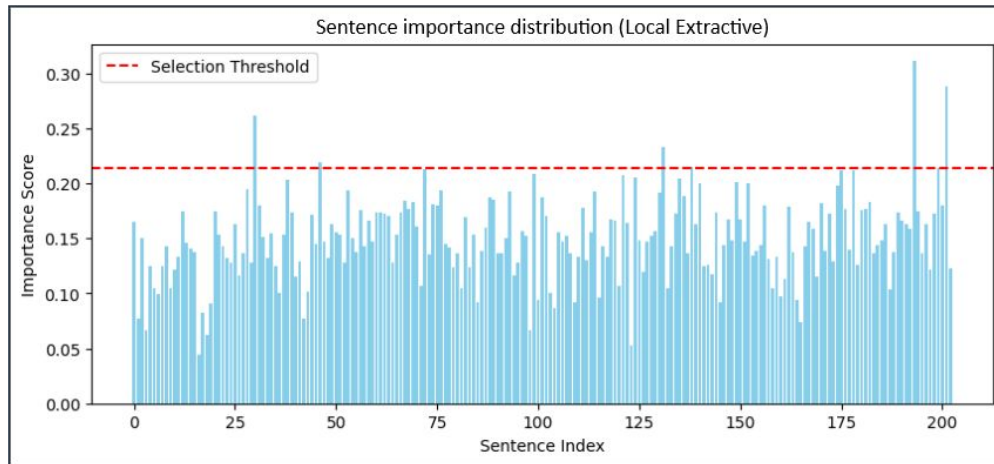
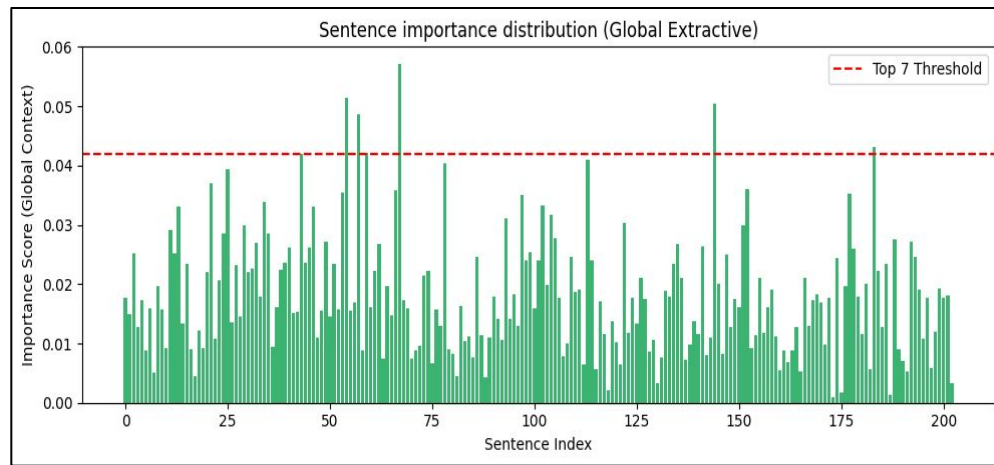
- Then we generate summaries from each document of the test set. To do so, we utilise both **extractive text summarisation** (global pLSA and local pLSA) and **abstractive text summarisation** (using a LED model and a LLM).
- Before performing text summarisation, we load and preprocess the **test set**, following the same steps we did earlier for the train set.



EXTRACTIVE SUMMARISATION

- This technique utilises matrices W and H obtained during topic modelling to **score the importance of each sentence** of a document, then the summary is generated by **selecting the highest-scoring sentences**.
- This technique is performed both **globally**, training the algorithm on the entire corpus before feeding it the test set, and **locally**, training it only on the current document.





- These are the results for **global** and **local** extractive summarisation. The graphs highlight the importance of each sentence in the first document: if it reaches over the threshold, then the corresponding sentence is selected to be part of the summary.



ABSTRACTIVE SUMMARISATION

- The **Longformer Encoder-Decoder (LED)** is a transformer optimised for long sequences; we select a version that's already fine-tuned for text summarisation tasks. This model uses a sliding-window based **attention mechanism** with a sparse attention mechanism to capture context.
- We utilise the **Large Language Model Qwen2.5-Instruct (7B)**, which is fine-tuned for following instructions. A **chunking and map-reduce** strategy must be implemented to fit the model weights without exceeding system memory.



TEXT SUMMARISATION EVALUATION

- **ROUGE** is an evaluation metric that considers exact matching of n-grams between summaries and reference summaries. We compute ROUGE-1, ROUGE-2 and ROUGE-L.
- **BERTScore** computes similarity using word embeddings, effectively capturing semantic similarity and overcoming abstraction.
- **SummaC** utilises a NLI model to verify the factual coherence of summaries and detect hallucinations.



RESULTS OF TEXT SUMMARISATION

- Results are as follows:

Technique	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore(F1)	SummaC
Global Extractive	0.23	0.05	0.20	0.8321	0.9437
Local Extractive	0.24	0.06	0.20	0.8316	0.9344
Abstractive (LED)	0.20	0.05	0.18	0.8175	0.8968
Abstractive (LLM)	0.26	0.07	0.23	0.8498	0.5089



**THANK YOU
FOR YOUR ATTENTION**

